

Kesavan G

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EXECUTIVE SUMMARY

As a final year B.Tech AI & Data Science student who loves building things that actually work. Able to analyse data and find patterns that matter, build and train machine learning models, create AI-powered applications, and develop Python backends — all the way from idea to deployment. Passionate about solving real problems with data and writing clean, practical code. Looking for an internship or entry-level role to contribute from day one and keep growing alongside a team.

EDUCATION

Ramco Institute of Technology

B.Tech - Artificial Intelligence and Data Science; Pre-Final Year

Rajapalayam, Tamil Nadu, India

May 2023 – May 2027

TECHNICAL SKILLS

Language : Python, MySQL, Java

ML & Data: Pandas, NumPy, Scikit-learn, XGBoost, CNN, RNN, LSTM, TensorFlow Lite, LLMs, RAG

Backend & Web: Flask, FastAPI, Streamlit, HTML, CSS, REST APIs

Tools & Infra: VS Code, Git, GitHub, Power BI, Canva, LiveKit

Focus Areas: Machine Learning, Data Analytics, Model Training, IoT Integration, Data Visualization, AI Applications

EXPERIENCE

Data Science Intern

Livewire

Jul 2025 – Aug 2025

Madurai, Tamil Nadu, India

- Analysed large eCommerce sales datasets using Python (Pandas, NumPy) — surfaced actionable KPIs and business trends.
- Built RNN model for multi-step time-series forecasting and predictive analytics on real business data.
- Delivered interactive dashboards with Matplotlib & Seaborn; used AI tools for automated insight generation.

Machine Learning Intern

Anjana Infotech

Feb 2025

Rajapalayam, Tamil Nadu, India

- Designed and evaluated ML pipelines end-to-end: preprocessing, feature engineering, model selection, and optimisation.
- Fine-tuned models for real-world business datasets — improving both prediction accuracy and runtime efficiency.

PROJECTS

Smart Home IoT Load Forecasting — Adaptive Hybrid Model

| Python, LSTM, XGBoost, Pandas, Scikit-learn

- Built an adaptive hybrid LSTM-XGBoost framework with behavioral feature engineering (load entropy, peak-hour indicators) for smart home energy forecasting.
- Achieved $R^2 = 0.9528$ and $RMSE = 0.1021$ using adaptive inverse-error ensemble weighting with 95% confidence interval uncertainty quantification.

Driver Drowsiness Detection — Edge AI

| Python, TensorFlow Lite, OpenCV, Jetson Nano

- Deployed real-time drowsiness detection on Jetson Nano using TF Lite — optimised for low-latency edge inference.
- Applied OpenCV for live video frame processing, eye-state classification, and on-device alert triggering

Friday — AI Voice Assistant

| Python, Gemini API, LiveKit, Google API

- Built a multilingual, real-time voice assistant with noise cancellation and optimised audio streaming pipelines.
- Integrated Gemini API for natural language responses; supports email automation, web search, and weather queries.

Compliance Checker — Cloud Policy Validator

| Python, Flask, Streamlit, Terraform, JSON

- Automated validation of Terraform configs against Google Cloud & CIS security benchmarks.
- Generated structured compliance reports — reducing manual cloud audit effort significantly.

Handwritten Records to Excel Automation

| Python, Flask, Google Gemini API, OpenPyXL, Unicorn, Render

- Built an AI web app that extracts data from handwritten/scanned records into formatted Excel sheets, cutting manual entry time from ~1 hour to under 1 minute.

Smart Agriculture Web – (Disease Detection, Yield Prediction, Smart Irrigation)

| Python, Flask (Ongoing)

- Developed an AI and IoT-powered agriculture platform integrating real-time sensor monitoring, NPK/soil data analysis, crop disease detection using YOLOv8 and EfficientNet, and a live dashboard using ESP32 and Supabase to support data-driven farming decisions.

CERTIFICATES / ACHIEVEMENTS

- **Kaggle – PhysioNet ECG Image Digitization:** Ranked top 300+ of 10,000+ participants, score 17+. Recovered from a -20 starting score by debugging the full ML pipeline — fixing preprocessing failures, silent model errors, and execution timeouts.
- **Deloitte Australia – Data Analytics Job Simulation (Forage):** Completed a virtual job simulation focused on **data analysis, data visualization, and business insights**, gaining practical experience in solving real-world analytics problems.
- **Tata – GenAI Powered Data Analytics Job Simulation (Forage):** Completed a simulation focused on applying **Generative AI in data analytics**, exploring AI-assisted analysis, data-driven decision-making, and business problem-solving
- **AI Tools Workshop – be10x:** Completed a hands-on workshop exploring **AI tools and their practical applications**, with a focus on improving productivity and leveraging AI for real-world tasks..
- **Reccsar – Power BI in Action Workshop: “Power BI in Action: Build, Analyze & Present Data,”** gaining hands-on experience with **DAX functions, data analysis, and creating meaningful interactive visualizations and dashboards.**